

Statistical Analysis of Customer Spending and Ordering Behavior at Starbucks

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Abstract. We analyze 100,000 Starbucks customer orders to quantify what drives the per-order spend and to test segment-level differences using the mathematical and statistical methods taught in DATA 202. The analysis combines descriptive statistics, Pearson correlation, Chebyshev’s inequality, distribution diagnostics, confidence intervals, two-sample and paired t -tests, two-proportion z -tests, empirical and Bayesian probability, a discrete random variable analysis of cart size, and a least-squares linear model solved through the normal equations and verified by gradient descent. Cart size is the dominant order-level driver of total spend with $r \approx 0.90$. Mobile App orders spend on average \$5.58 more than non-mobile orders, and after controlling for cart size and customizations the Mobile premium is still \$2.33. Rewards membership shifts spend up by a smaller \$1.63. The fitted linear model achieves $R^2 \approx 0.95$ on the training sample. We discuss how the same analysis pipeline could be deployed inside a Starbucks marketing or revenue-management workflow.

I. INTRODUCTION

Starbucks operates one of the largest retail coffee networks in the world. Understanding what drives per-order revenue at a single transaction is central to merchandising, mobile-channel investment, and rewards-program design. The DATA 202 syllabus covers measurements and sampling, descriptive statistics, correlation and covariance, distribution theorems, confidence intervals, hypothesis testing, probability and random variables, distributions, calculus and optimization, and linear algebra [2]. This report uses each of those tools on a single open dataset to answer one practical question: *do customers spend more in certain channels or segments, and what variables actually move the per-order spend?*

The contribution is methodological rather than novel. We treat the analysis as a complete worked example of the course content applied to a 100,000-row real-world dataset, with every formula written out, computed, and visualized. Sections are organized so that each block of analysis maps directly to one or two course lectures.

II. DATA DESCRIPTION

A. Source

The dataset is the public *Starbucks Customer Ordering Patterns* table on Kaggle [1]. Each row is a single customer order. The file contains 100,000 orders placed by 14,988

TABLE I
DESCRIPTIVE STATISTICS FOR QUANTITATIVE VARIABLES

Variable	Mean	Median	SD	IQR
total_spend (\$)	14.87	14.17	5.51	7.34
cart_size	3.74	4.00	1.70	2.00
num_customizations	1.81	2.00	1.46	2.00
fulfillment_time_min	4.55	4.40	1.55	2.10
customer_satisfaction	3.69	4.00	1.18	2.00

unique customers across 500 stores, with no missing values, between 2024-01-01 and 2025-12-30. There are 20 columns covering customer attributes, store attributes, channel, time, cart size, customizations, total spend, fulfillment time, drink category, food attachment, order-ahead flag, and customer satisfaction.

B. Variables

The 20 columns split into qualitative and quantitative variables. Qualitative columns include `customer_id`, `order_id`, `order_channel`, `store_location_type`, `region`, `customer_age_group`, `customer_gender`, `drink_category`, and three booleans (`is_rewards_member`, `has_food_item`, `order_ahead`). Quantitative columns are `cart_size`, `num_customizations`, `total_spend`, `fulfillment_time_min`, and `customer_satisfaction` (treated as ordinal). The response variable in this study is `total_spend`.

C. Descriptive Statistics

For the five quantitative columns we compute the mean, median, mode, standard deviation, variance, and the five-number summary. Table I reports the values that drive the rest of the report. The mean and median of `total_spend` are close (\$14.87 vs \$14.17) and the standard deviation is \$5.51, which is consistent with a roughly symmetric distribution with a slight right skew. The IQR fences flag only 1.42% of orders as outliers on `total_spend`; we keep them.

D. Channel and Segment Mix

About 42.5% of orders are placed through the Mobile App, 28.0% Drive-Thru, 22.1% In-Store Cashier, and 7.4%

Kiosk. Rewards members account for 47.7% of orders. Region, store location type, and drink category are roughly evenly distributed. The largest single segmentation effect on mean spend comes from the order channel: Mobile App orders average \$18.08 while the other three channels are tightly clustered near \$12.50.

III. MATHEMATICAL AND STATISTICAL METHODS

This section states the formulas used in the rest of the paper. Each formula is taken directly from a DATA 202 lecture.

A. Pearson Correlation and Sample Covariance

For paired observations $(x_i, y_i)_{i=1}^n$, sample covariance and Pearson correlation are

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad (1)$$

$$r_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}. \quad (2)$$

B. Chebyshev's Inequality and the Empirical Rule

Chebyshev's inequality holds for any distribution with finite mean and variance:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}. \quad (3)$$

The empirical (68/95/99.7) rule is a tighter approximation that requires X to be close to normal.

C. Confidence Interval for a Mean

The two-sided $100(1 - \alpha)\%$ confidence interval for μ from a sample of size n is

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}. \quad (4)$$

For the sample sizes in this study ($n > 10,000$) the t -quantile is indistinguishable from $z = 1.96$ at $\alpha = 0.05$.

D. Welch's Two-Sample t -test

For two independent samples with sizes n_1, n_2 , sample means $\bar{x}_1, \bar{x}_2$, and variances s_1^2, s_2^2 , the Welch statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}. \quad (5)$$

E. Paired t -test

For matched pairs with within-pair difference d_i , the test statistic is

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}, \quad H_0 : \mu_d = 0. \quad (6)$$

F. Two-Proportion z -test

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}, \quad \hat{p} = \frac{x_1 + x_2}{n_1 + n_2}. \quad (7)$$

G. Empirical Probability, Conditional Probability, Bayes

$$\hat{P}(A) = \frac{\#\{i : x_i \in A\}}{n}, \quad P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad (8)$$

$$P(A | B) = \frac{P(B | A) P(A)}{P(B | A) P(A) + P(B | A^c) P(A^c)}. \quad (9)$$

H. Discrete Random Variable Moments

$$E[X] = \sum_k k \hat{p}(k), \quad \text{Var}(X) = \sum_k (k - E[X])^2 \hat{p}(k). \quad (10)$$

I. Least Squares: Normal Equations and Gradient Descent

With $y \in \mathbb{R}^n$, design matrix $X \in \mathbb{R}^{n \times p}$, and squared-error loss $J(\beta) = \|y - X\beta\|^2$, setting $\nabla_\beta J = 0$ gives the normal equations

$$X^\top X \beta = X^\top y, \quad \hat{\beta} = (X^\top X)^{-1} X^\top y. \quad (11)$$

The same minimizer can be reached iteratively from

$$\beta_{t+1} = \beta_t - \eta \nabla_\beta J(\beta_t), \quad \nabla_\beta J(\beta) = -2X^\top (y - X\beta). \quad (12)$$

IV. ANALYSIS AND RESULTS

A. Order-Level Correlation and Covariance

We compute (1) and (2) on the order-level numerical block. Cart size is the dominant driver of `total_spend` with $r = 0.900$, followed by `num_customizations` ($r = 0.409$). Fulfillment time and customer satisfaction are essentially uncorrelated with `spend` ($|r| < 0.07$). Manual computation reproduces the pandas value to four decimal places: $\text{Cov}(\text{cart_size}, \text{total_spend}) = 8.4159$ and $r = 0.9001$. Figure 1 shows the order-level correlation heatmap and Figure 2 shows the two predictor scatter plots that drive the model in Section IV-G.

A customer-level aggregation was also computed for documentation but not used as the headline. Summing customizations within a customer inflates correlation with summed `spend` (from $r=0.41$ at the order level to $r=0.81$ at the customer level), and averaging cart size deflates it ($r=0.90$ to $r=0.30$). Both effects are sum-of-sums or averaging artifacts. The order-level numbers are the right ones to interpret.

B. Distribution of Total Spend

Applying (3) to `total_spend` with $\mu = 14.87$ and $\sigma = 5.51$ yields the bounds in Table II. The actual within-band shares are far above Chebyshev's worst case and very close to the empirical-rule values, indicating that the distribution is much closer to normal than the worst case. Figure 3 overlays the $\pm 2\sigma$ and $\pm 3\sigma$ bands on the histogram.

A formal normality check (Figure 4) shows skewness 0.6701, excess kurtosis 0.3775, and a Shapiro-Wilk statistic of $W = 0.9734$ ($p \approx 1.6 \times 10^{-29}$) on a random subsample of size 5,000. The body of the distribution is approximately normal but the right tail is heavier than normal. We therefore use empirical probabilities for tail events instead of the normal CDF.

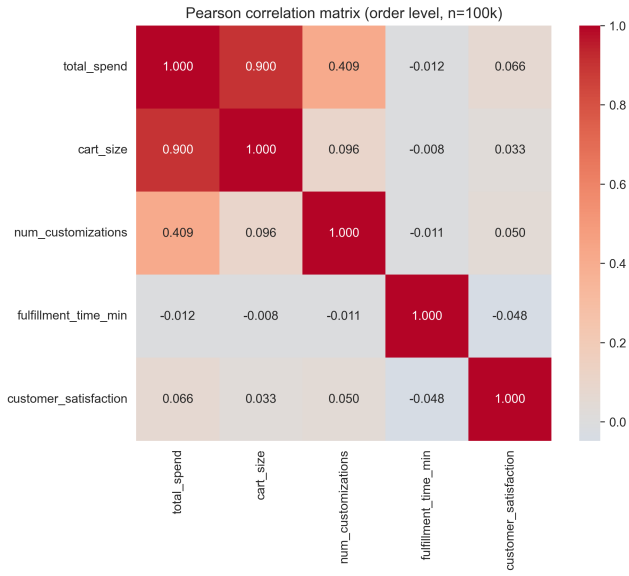


Fig. 1. Pearson correlation matrix at the order level ($n = 100,000$).

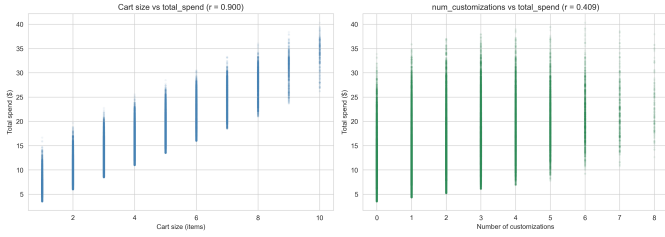


Fig. 2. Order-level scatter plots of cart_size and num_customizations against total_spend.

TABLE II
CHEBYSHEV VS EMPIRICAL RULE VS ACTUAL FOR TOTAL_SPEND

k	Chebyshev min.	Empirical rule	Actual
2	75.00%	95.4%	95.83%
3	88.89%	99.7%	99.34%

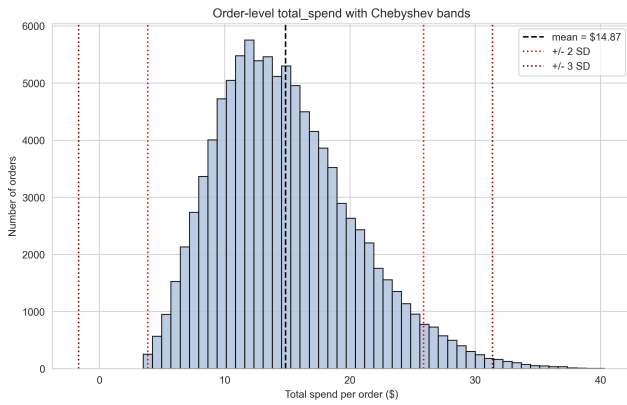


Fig. 3. Histogram of total_spend with mean and $\pm 2\sigma, \pm 3\sigma$ bands.

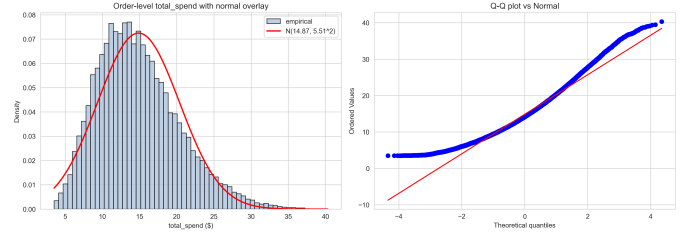


Fig. 4. Histogram with normal overlay (left) and Q-Q plot vs normal (right) for total_spend.

TABLE III
SELECTED 95% CONFIDENCE INTERVALS FOR MEAN TOTAL_SPEND

Segment	n	Mean (\$)	95% CI (\$)
Mobile App	42,521	18.08	[18.02, 18.13]
Drive-Thru	27,996	12.48	[12.43, 12.53]
In-Store Cashier	22,063	12.51	[12.46, 12.57]
Kiosk	7,420	12.49	[12.40, 12.58]
Rewards member	47,717	15.72	[15.67, 15.77]
Non-member	52,283	14.09	[14.04, 14.13]
Age 18-24	20,274	15.55	[15.47, 15.63]
Age 55+	10,019	13.06	[12.97, 13.15]

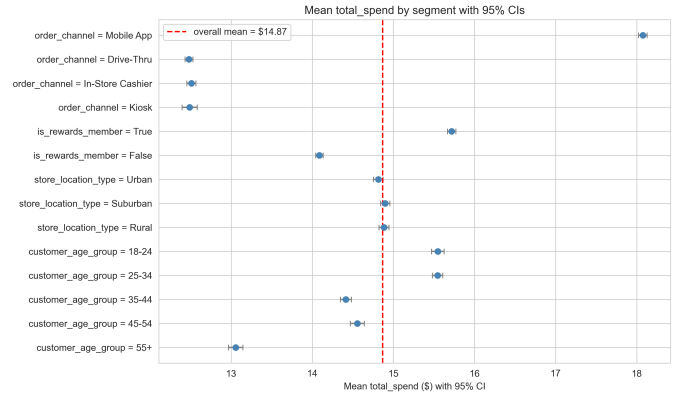


Fig. 5. Mean total_spend with 95% confidence intervals by segment. The dashed red line is the overall mean.

C. Confidence Intervals

Applying (4) at $\alpha = 0.05$ gives an overall 95% CI for the mean spend of $[\$14.83, \$14.90]$. Table III reports CIs by segment and Figure 5 visualizes them. The Mobile App segment is clearly separated from the other channels, and the rewards/non-rewards CIs do not overlap.

D. Hypothesis Tests

We run four hypothesis tests at $\alpha = 0.05$.

Mobile vs non-Mobile. With $H_0 : \mu_1 = \mu_2$ and $H_a : \mu_1 \neq \mu_2$, (5) gives $\bar{x}_1 - \bar{x}_2 = \5.58 , $t = 175.91$, and $p \approx 0$. The 95% CI for the difference is $[\$5.52, \$5.65]$. We reject H_0 .

Rewards vs non-rewards. The same procedure gives $\bar{x}_1 - \bar{x}_2 = \1.63 , $t = 47.08$, $p \approx 0$, and CI $[\$1.56, \$1.70]$. We reject H_0 . The effect is small in dollars but the sample is large enough to detect it.

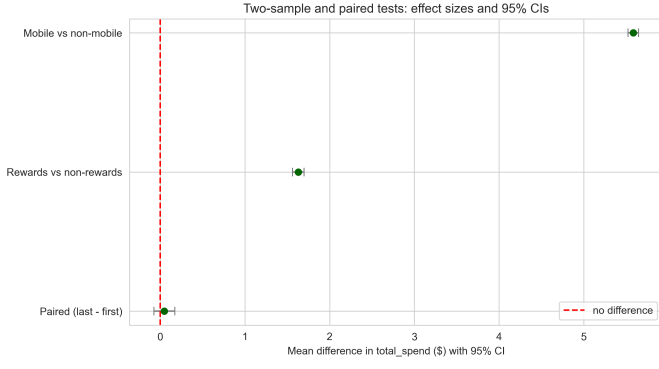


Fig. 6. Mean differences in total_spend with 95% CIs. The dashed red line is the no-difference reference.

TABLE IV
EMPIRICAL VS NORMAL-APPROXIMATION $P(\text{TOTAL_SPEND} > t)$

t (\$)	$\hat{P}_{\text{emp}}$	$\hat{P}_{\text{normal}}$	Difference
10	0.8054	0.8116	-0.006
15	0.4413	0.4904	-0.049
20	0.1718	0.1756	-0.004
25	0.0511	0.0329	+0.018
30	0.0107	0.0030	+0.008

Paired test, first vs latest order. For the 14,857 customers with at least two orders, (6) gives $\bar{d} = \$0.05$, $t = 0.77$, $p = 0.44$, CI $[-\$0.07, \$0.17]$. We fail to reject H_0 . There is no within-customer drift in spend over the 24-month window.

High-spender proportions. Defining a high-spender order as one with $\text{total_spend} > \$20$ and applying (7), the rate is 0.2167 for rewards and 0.1309 for non-rewards. The pooled $z = 35.92$, $p \approx 0$, and the unpooled 95% CI for the difference is $[0.0811, 0.0905]$. We reject H_0 . Rewards membership is associated with a strictly higher tail rate, consistent with the mean test.

Figure 6 summarizes the three mean-difference tests on a single axis.

E. Empirical Probability and Bayes

Empirical tail probabilities computed via (8) appear in Table IV. Figure 7 compares the empirical curve with a normal approximation that uses the sample μ and σ . The normal approximation overstates the share near the median (gap -0.049 at \$15) and understates the right tail (gap $+0.018$ at \$25, $+0.008$ at \$30, in absolute terms; in relative terms the underestimate is roughly 35% and 72%). For tail questions we use the empirical numbers.

Conditional probabilities of a high-spender order ($> \$25$) are shown in Figure 8. By channel, $P(\text{spend} > 25 \mid \text{Mobile}) = 0.1127$ versus about 0.005 for the other three channels, a $\approx 20\times$ ratio. By rewards, $P(\text{spend} > 25 \mid \text{Rewards}) = 0.0672$ versus 0.0364, a $\approx 1.85\times$ ratio.

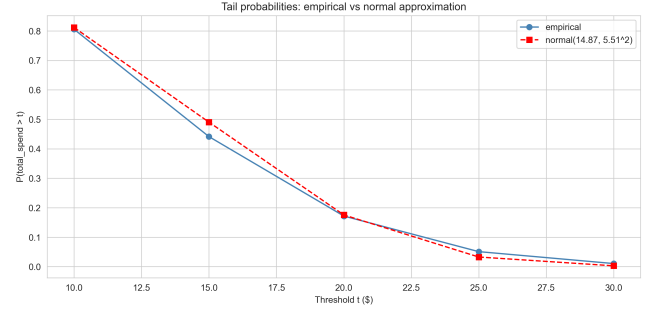


Fig. 7. Tail probabilities $P(\text{total_spend} > t)$: empirical vs normal approximation.

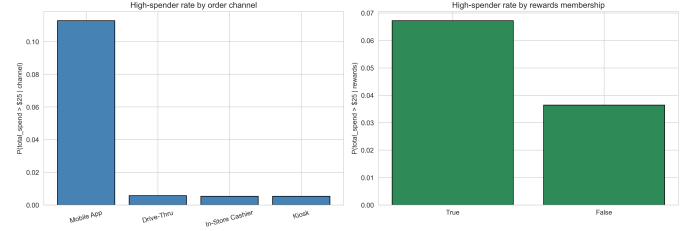


Fig. 8. Conditional high-spender probabilities by channel and rewards membership.

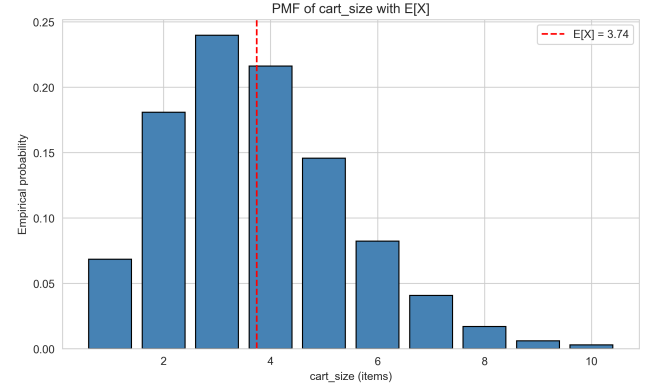


Fig. 9. Empirical PMF of cart_size with $E[X]$ marked.

Applying Bayes' theorem (9) with $A = \{\text{Mobile App}\}$ and $B = \{\text{spend} > 25\}$:

$$P(A \mid B) = \frac{0.1127 \cdot 0.4252}{0.1127 \cdot 0.4252 + 0.0055 \cdot 0.5748} = 0.9385.$$

A direct count gives 0.9385, confirming the chain. Almost all high-spender orders come from the Mobile App. The same procedure on rewards gives $P(\text{Rewards} \mid \text{spend} > 25) = 0.6274$.

F. Random Variable: Cart Size

cart_size is a discrete random variable on $\{1, \dots, 10\}$. The empirical PMF (Figure 9) is concentrated on 3 to 5 with a right tail to 10. Applying (10) gives $E[X] = 3.7415$ and $\text{Var}(X) = 2.8826$, matching the descriptive moments in Table I.

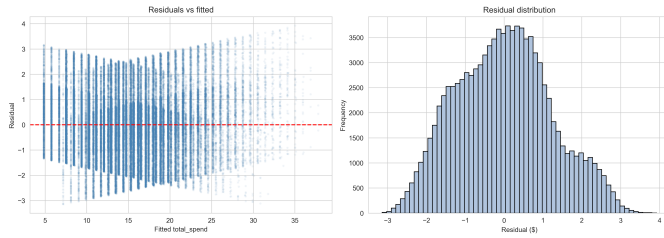


Fig. 10. Linear model diagnostics: residuals vs fitted values (left) and residual histogram (right).

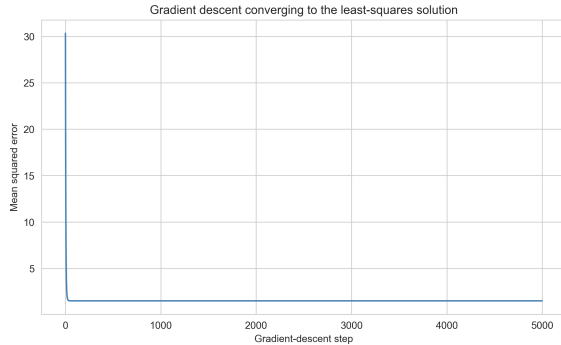


Fig. 11. Gradient descent on the standardized least-squares loss converges to the closed-form solution.

G. Linear Model via Normal Equations and Gradient Descent

We fit

$$y_i = \beta_0 + \beta_1 \text{cart}_i + \beta_2 \text{custom}_i + \beta_3 \mathbf{1}\{\text{Mobile}_i\} + \varepsilon_i.$$

The design matrix is $X \in \mathbb{R}^{100000 \times 4}$ with a leading column of ones. Solving the normal equations (11) gives

$$\hat{\beta} = (2.156, 2.685, 0.924, 2.331).$$

The condition number of $X^\top X$ is 1.54×10^2 , well within the safe range. The model fit summary on the full sample is $\text{RSS} = 150,098$, $\text{TSS} = 3,032,454$, $R^2 = 0.9505$, and residual standard error $\hat{\sigma} = \$1.23$. Diagnostics in Figure 10 are well-behaved with a roughly bell-shaped residual distribution and no obvious systematic pattern.

Optimization view. Standardizing the predictors and running gradient descent (12) from $\beta = 0$ at learning rate 0.05 converges to the same coefficients in 5,000 steps, with L2 distance 0 to the closed-form solution. Figure 11 shows the loss curve. This connects Lecture 9 (calculus and optimization) to Lectures 10 and 11 (linear algebra and systems of linear equations).

V. DISCUSSION AND INTERPRETATION

A. What the Numbers Mean for the Business

The Mobile App channel is the single largest revenue lever in the data. Mobile orders average \$5.58 more than non-Mobile orders, and even after controlling for cart size and customizations the channel premium is still \$2.33 per

order. The conditional and Bayesian analysis tells the same story from a different angle: about 94% of orders above \$25 are placed through the Mobile App. Rewards membership matters but in a smaller way (\$1.63 per order on average, $1.85\times$ higher tail rate). Within a customer's history, the first and latest order do not differ on average, so observed cross-segment differences are between-segment effects rather than within-customer drift.

B. Cross-Method Consistency

The same conclusions show up under multiple methods. The mean test, the proportion test on the right tail, and the Bayesian posterior all point to Mobile being the dominant high-spender segment. The linear model splits credit between cart size, customizations, and Mobile in a way that is consistent with the order-level correlations: the strongest predictor by correlation also has the largest standardized coefficient.

C. Deployment

Translating the analysis into a production setting at Starbucks scale would proceed in three layers. (i) *Batch refresh*: a daily or hourly job recomputes the descriptive statistics, segment means and CIs, and updates the linear model coefficients. The closed-form $\hat{\beta} = (X^\top X)^{-1} X^\top y$ is cheap when p is small and n is large because $X^\top X \in \mathbb{R}^{p \times p}$ does not grow with n . (ii) *Online scoring*: at the point of sale, given a customer's cart, customization count, and channel flag, the model produces a one-step expected-spend prediction in microseconds, suitable for in-app upsell or rewards-aware promotions. (iii) *Monitoring*: the empirical distribution of spend, segment proportions, and conditional probabilities should be tracked over time. Drift in $P(\text{spend} > 25 \mid \text{Mobile})$ or in the Mobile-vs-others gap would trigger a model refit. Scalability is handled by partitioning over store or region, since the model is linear and additive in segment indicators. Dynamic-environment risks include channel mix shifts (e.g., Mobile share growing further) and rewards-program changes; both are captured by recomputing the segment statistics on the freshest window. The linear model is intentionally simple and is meant as a course-aligned baseline rather than a production predictor.

D. Limitations

- The dataset is a single Kaggle file and may not represent all Starbucks stores or regions.
- Customers appear in multiple orders, so strict independence across rows is imperfect. Welch's test and the empirical probabilities still apply, but standard errors are slightly optimistic.
- The Shapiro-Wilk normality test was run on a random sub-sample of 5,000 orders because the test is unreliable on $n = 100,000$. The conclusion (rejected at any reasonable level) would not change with a larger sub-sample.
- The normal-CDF approximation underestimates right-tail spend, which is why the empirical numbers are reported in Section IV-E.

- The paired first-vs-latest test ignores the unequal time gaps between a customer’s first and latest order; it only checks whether there is any net within-customer drift on average.
- The linear model is intentionally simple and course-aligned. It does not include time, region, store, or interaction terms, and its $R^2=0.95$ comes mostly from the very strong cart-size correlation.

VI. CONCLUSION

We applied descriptive statistics, correlation, distribution diagnostics, confidence intervals, hypothesis testing, empirical and Bayesian probability, discrete random variable analysis, and least-squares regression with both closed-form and gradient-descent solutions to a 100,000-order Starbucks dataset. The Mobile App channel is the dominant revenue lever; cart size is the dominant order-level numerical predictor. Rewards membership is a smaller but statistically significant lever. All findings are consistent across the methods. The same pipeline can be deployed as a daily batch job feeding an online scoring layer at the point of sale, with monitoring of empirical segment statistics to catch drift.

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